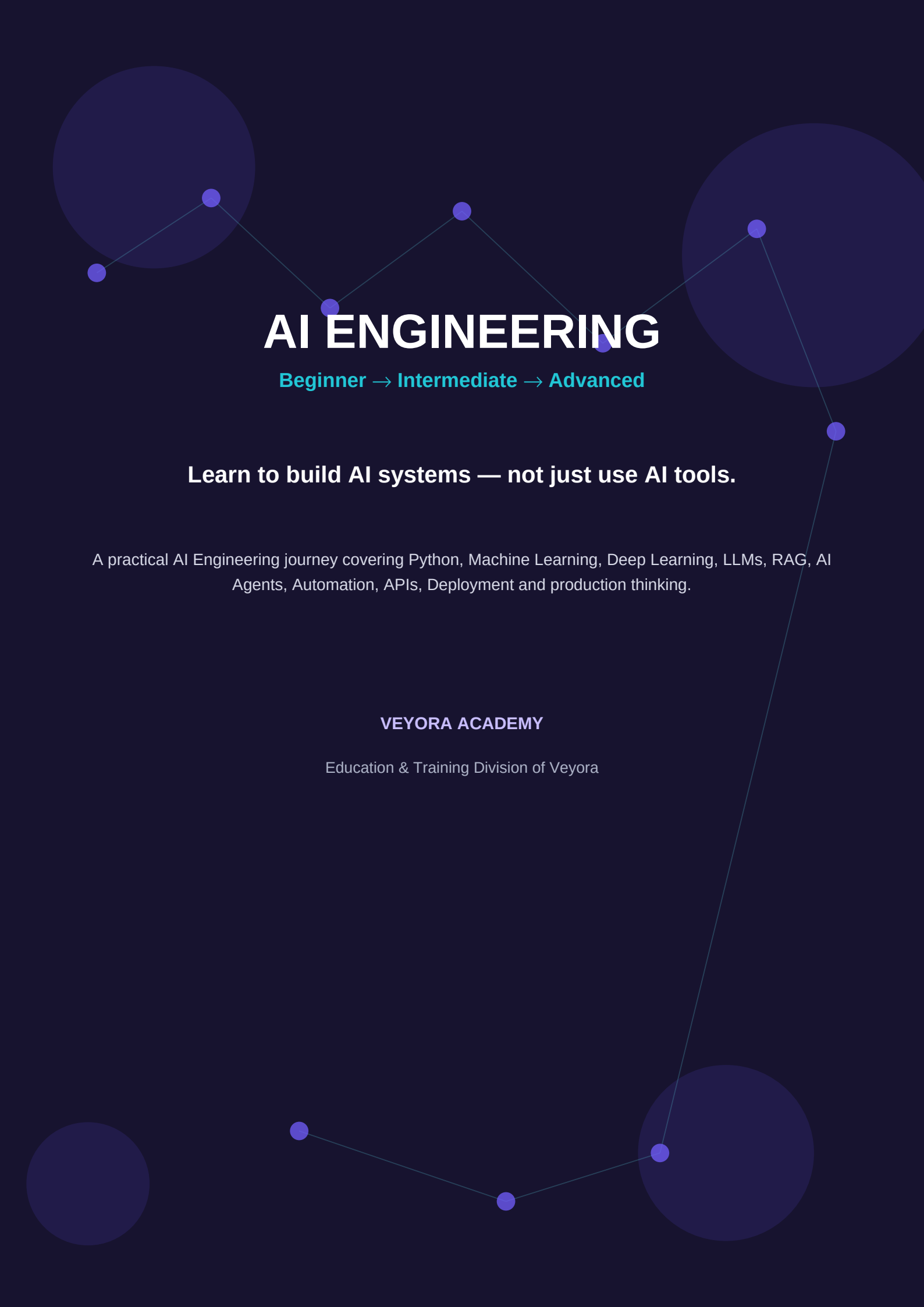




AI ENGINEERING

Beginner → Intermediate → Advanced

Learn to build AI systems — not just use AI tools.

A practical AI Engineering journey covering Python, Machine Learning, Deep Learning, LLMs, RAG, AI Agents, Automation, APIs, Deployment and production thinking.

VEYORA ACADEMY

Education & Training Division of Veyora

Why AI Engineering?



- AI is moving from simple chatbots and prompts into real software systems.
- AI Engineers build intelligent applications, recommendation systems, document-processing systems, copilots, AI agents and automated workflows.
- This program is designed to take a learner from programming fundamentals to production-oriented AI systems.

What You Will Learn & Build

Core AI Engineering Stack

- Python & Software Engineering
- Data & Machine Learning
- Deep Learning
- LLM Engineering
- RAG Systems
- AI Agents
- AI Automation
- Deployment & Production AI

Learning Journey

- 1. Programming Foundation** — Python, Git, APIs, JSON, software engineering
- 2. Data + Machine Learning** — NumPy, Pandas, visualization, ML algorithms, evaluation
- 3. Deep Learning** — Neural networks, training, embeddings, computer vision concepts
- 4. LLM Engineering** — LLM APIs, prompting, structured outputs, function/tool calling
- 5. RAG + AI Applications** — Embeddings, vector search, retrieval, document AI, evaluation
- 6. AI Agents + Automation** — Agent architecture, tools, memory, workflows, multi-step systems
- 7. Advanced AI Systems** — Fine-tuning concepts, multimodal AI, evaluation, safety
- 8. Deployment + MLOps** — APIs, containers, testing, monitoring, production thinking

Beginner Level



Programming & Developer Foundations

- Python variables, data types, conditions, loops and functions
- Lists, dictionaries, modules and exceptions
- VS Code, Git, GitHub, virtual environments, packages and debugging
- CSV, JSON, files, APIs and basic SQL
- Practical AI mathematics: probability, statistics, vectors, matrices and gradients

Intermediate Level

Data + Machine Learning

- NumPy, Pandas, data cleaning, missing values and feature preparation
- Matplotlib and exploratory data analysis (EDA)
- Regression, classification and clustering
- Train/test split and model evaluation
- Accuracy, precision, recall, F1 score and confusion matrix
- Overfitting and model improvement
- ML workflow: Problem → Data → Features → Model → Evaluation → Improvement

Advanced Level

Deep Learning

- Neurons, layers, activations, loss functions, backpropagation and training
- Embeddings and representation learning
- Transfer learning
- Computer vision and OCR concepts

LLM Engineering

- Tokens, context and model behavior
- Prompt engineering and system instructions
- Structured outputs
- Function and tool calling
- Model APIs and streaming
- Cost and latency considerations

Advanced AI Systems



RAG — Retrieval-Augmented Generation

- Document ingestion and preprocessing
- Chunking strategies
- Embeddings and vector databases
- Semantic search and retrieval
- Context construction and grounded answers
- RAG evaluation

AI Agents

- Agent architecture
- Tool use and API interaction
- Planning and multi-step execution
- Memory concepts
- Workflow orchestration
- Human-in-the-loop systems
- Agent evaluation

Production AI

- AI automation and multimodal AI
- Fine-tuning concepts
- AI evaluation and safety
- Secure API handling and access control
- APIs and deployment
- MLOps concepts
- Reliability, latency, scalability and observability
- Fallbacks and AI product UX

Tools & Technologies

Programming	Python • VS Code • Git • GitHub • Jupyter
Data	NumPy • Pandas • Matplotlib • SQL
Machine Learning	Scikit-learn • PyTorch / TensorFlow concepts
GenAI	OpenAI APIs • Other model APIs • Embeddings • Vector databases
Automation	n8n • Make • APIs • Workflow orchestration
Applications	FastAPI • REST APIs • Streamlit or similar interfaces
Production	Docker concepts • Cloud deployment concepts • Monitoring

Practical Projects

- Python Automation Project
- Data Analysis + Machine Learning Project
- AI Prediction Application
- LLM Application
- RAG Knowledge Assistant
- AI Agent
- AI Automation System
- Final Capstone — a production-oriented AI system



Career Directions

- AI Engineer
- Machine Learning Engineer
- LLM / Generative AI Engineer
- AI Automation Engineer
- AI Application Developer
- Data + AI roles
- Freelancing and client AI solution work

Student Requirements

- Laptop and stable internet connection
- Basic computer literacy and willingness to practice
- Strong mathematics, prior programming and data experience are helpful but not mandatory

Certificate

Students who complete the required learning, practical work and assessment criteria can receive a Veyora Academy Certificate.

VEYORA — BUSINESS & ACADEMY INFORMATION

Veyora: Registered Technology & Digital Solutions Business

Tax Registration: NTN / FBR Registered Business

Veyora Academy: Official Education & Training Division of Veyora

Student Certification: Veyora Academy Certificates are issued upon successful completion of the applicable program requirements.

Responsible AI

The program also emphasizes responsible AI practices, including data privacy, access control, secure API handling, evaluation, hallucination awareness, prompt-injection awareness and appropriate human oversight.

Veyora Academy is the education and training division of Veyora, a registered technology and digital solutions business.